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Calculation of High-Pressure Vapor-Liquid Equilibria from a Corresponding-States Correlation with Emphasis on Asymmetric Mixtures

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The correlation of Lee and Kesler for thermodynamic properties of normal fluids is applied to mixtures. Vapor-liquid equilibria at high pressures are calculated using a one-fluid theory with a new mixing rule for the pseudo-critical temperature; this mixing rule appears to be applicable also to those mixtures whose components differ appreciably in molecular size. The calculational procedure, using analytical functions, is completely computerized; computer programs are available. Input data are pure-component critical properties and acentric factors plus one binary constant for each binary pair. These constants are tabulated for a large number of binary systems containing light gases (including hydrogen), carbon dioxide, hydrogen sulfide, and hydrocarbons ranging from methane to decane. Illustrative calculations indicate that the procedure described here provides good estimates of K factors, enthalpies, and other thermodynamic properties required in engineering process design.

The configurational properties of normal fluids were correlated by Pitzer and co-workers 20 years ago through an extended theorem of corresponding states which, in addition to critical temperature and critical pressure, utilized the acentric factor as a third correlating parameter (Pitzer et al., 1955). Although the correlation was based exclusively on pure-component properties, extension to mixtures is possible through the pseudo-critical concept of Kay (1936) as modified by various authors (Prausnitz and Gunn, 1958; Pitzer and Hultgren, 1958; Leland and Mueller, 1959; Barner and Quinlan, 1969). However, since Pitzer's results were presented in tabular form, they are not convenient for calculating phase equilibria; such calculations are unavoidably of the trial-and-error type and therefore, since electronic computers must be used, analytical representation of configurational properties is highly desirable.

Lee and Kesler (1975) established a correlation similar to Pitzer's; utilizing experimental data not available 20 years ago, Lee and Kesler's correlation is applicable to temperatures lower than those in Pitzer's tables. More important, however, Lee and Kesler present their results not only in tabular form but also through a reduced equation of state, thereby facilitating computer calculations.

In this work we discuss the application of that reduced equation to the calculation of vapor-liquid equilibria and enthalpies of multicomponent mixtures. Our discussion is similar to that of Joffe (1976), but it applies to a much wider variety of mixtures. Because of the increasing importance of

mixtures encountered in processing of synthetic gas from coal and heavy petroleum feed stocks, we give particular attention to asymmetric mixtures, that is, those which contain both small molecules (e.g., hydrogen, nitrogen, and methane) and large molecules (e.g., high-boiling paraffins, aromatics, and fused-ring hydrocarbons). These mixtures, which are becoming increasingly important in contemporary technology, are often not well represented by other methods for calculating vapor-liquid equilibria because conventional techniques are usually restricted to mixtures where differences in molecular size are not large.

Correlation of Lee and Kesler

The thermodynamic properties of normal fluids are expressed by a modified Benedict-Webb-Rubin (1940) equation. The equation of state is written first for a simple fluid (0), such as argon or methane, and then for a reference fluid (r), chosen to be *n*-octane. The compressibility factor z for a normal fluid, at reduced temperature T_r , reduced volume v_r , and acentric factor ω is written in the form

$$z(T_r, v_r, \omega) = z^{(0)}(T_r, v_r, \omega = 0) + \frac{\omega}{\omega^{(r)}} [z^{(r)}(T_r, v_r, \omega^{(r)}) - z^{(0)}(T_r, v_r, \omega = 0)] \quad (1)$$

The reduced variables are defined by

$$T_r = T/T_c; \quad v_r = \frac{p_c \cdot v}{R \cdot T_c} \quad (2)$$

The compressibility factor z is conveniently written

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$$z = \frac{p_r \cdot v_r}{T_r} \quad (\text{where } p_r = p/p_c) \quad (3)$$

The subscript c denotes critical. The pertinent equations for calculating compressibility factors, fugacities, enthalpies, and entropies are given in Appendix A.

Thermodynamic Framework

We consider a multicomponent, two-phase system at temperature T and total pressure p . The condition of equilibrium is that for every component i

$$f_i^V = f_i^L \quad (4)$$

where f stands for fugacity and superscripts V and L designate, respectively, the vapor phase and the liquid phase.

Following conventional procedure, we define the fugacity coefficient ϕ_i by

$$\phi_i = f_i/Z_i p \quad (5)$$

where Z_i is the mole fraction either in the liquid phase or in the vapor phase.

As discussed elsewhere (Prausnitz, 1969; van Ness, 1955), the fugacity coefficient for component i can be obtained by differentiating ϕ_M , the fugacity coefficient of the mixture, according to

$$\ln \phi_i = \ln \phi_M - \sum_{j \neq i} Z_j \left(\frac{\partial \ln \phi_M}{\partial Z_j} \right)_{Z_k} \quad (k \neq i, j) \quad (6)$$

where $\ln \phi_M$ is a function depending on T/T_{cM} , p/p_{cM} and ω_M . Here for each phase T_{cM} and p_{cM} are, respectively, the pseudo-critical temperature and pseudo-critical pressure and ω_M is the acentric factor. Appendix A presents the final equations.

The residual enthalpy of the mixture is

$$\Delta h_M = h(T, p, Z) - h(T, p = 0, Z) \quad (7)$$

Equations 6 and 7 are applied separately to each phase, once to the liquid phase and once to the vapor phase. The fugacity coefficient ϕ_M is also calculated separately, once for the liquid phase and once for the vapor phase, according to

$$\ln \phi_M = \frac{1}{RT} \times \int_V^\infty \left[\frac{p}{n_T} - \frac{RT}{V} \right] dV - \ln z_M + (z_M - 1) \quad (8)$$

where V is the total volume containing n_T number of moles and z is the compressibility factor ($z_M = pV/n_T RT$).

To calculate ϕ_M , z_M , and Δh_M we use a one-fluid theory; that is, we assume that the residual properties of a uni-phase mixture are those found from an equation of state for a hypothetical pure fluid whose characteristic constants are found by suitably averaging the constants of the pure components comprising the mixture. The averaging procedure is determined by essentially empirical mixing rules which, once established, also determine the derivatives with respect to Z appearing in eq 6.

The mixing rules proposed here are

$$T_{cM} = \frac{1}{v_{cM}^\eta} \cdot \sum_j \sum_k Z_j Z_k \cdot v_{cjk}^\eta \cdot T_{cjk} \quad (9)$$

$$v_{cM} = \sum_j \sum_k Z_j Z_k \cdot v_{cjk} \quad (10)$$

$$\omega_M = \sum_j Z_j \cdot \omega_j \quad (11)$$

where v_c is the molar critical volume, T_c is the critical temperature, and ω is the acentric factor. The cross coefficients are given by

$$T_{cjk} = (T_{cj} \cdot T_{ck})^{1/2} \cdot \kappa_{jk} \quad (12)$$

and

$$v_{cjk} = \frac{1}{8} (v_{cj}^{1/3} + v_{ck}^{1/3})^3 \quad (13)$$

where κ_{jk} is an adjustable binary parameter, characteristic of the j - k binary, independent of temperature, density, and composition. The pseudo-critical pressure is found by

$$p_{cM} = (0.2905 - 0.085\omega_M) \cdot R \cdot T_{cM}/v_{cM} \quad (14)$$

The mixture rules above are similar to those presented previously (Prausnitz and Gunn, 1958; Barner and Quinlan, 1969; Reed and Gubbins, 1973; Lee and Kesler, 1975) but contain one important variation: the exponent η . If we set $\eta = 0$ we obtain mixing rules similar to those of Prausnitz and Gunn (1958) which were also used by Barner and Quinlan (1969). If we set $\eta = 1$ we obtain (essentially) the so-called van der Waals mixing rules, used by Leland et al. (1968) and by Molterup (1975).

For symmetric mixtures ($v_{cj} = v_{ck}$), the exponent η is of no importance and thus η may be conveniently set equal to zero. For slightly asymmetric mixtures, Rowlinson and others have shown that setting $\eta = 1$ gives better agreement with the computer results of Singer (1970) as well as with experiment. However, for strongly asymmetric mixtures we find that while $\eta = 0$ does not take the effect of molecular size into proper account, setting $\eta = 1$ tends to overestimate that effect. We have therefore determined η empirically by a systematic study of numerous mixtures of normal fluids whose molecular size differ appreciably, that is, by a factor 3 or more. We find that while the optimum value of η varies somewhat from one system to another, good agreement with experiment is obtained by setting $\eta = 0.25$ for all binary systems considered here. To illustrate, Figure 1 shows deviations between calculated and experimental pressures for the two-phase system methane- n -heptane at 310.93 K for different values of η . For each calculation an optimum value of κ_{ij} was determined from the experimental data of Reamer et al. (1956).

Figure 2 presents deviations between calculated and observed bubble pressures for different values of κ_{ij} , again for the system methane- n -heptane at 310.93 K with $\eta = 0.25$. It is evident that results are sensitive to the value of κ_{ij} . Whereas a universal, not optimized value of η tends to give both negative and positive deviations in pressure, changing the value of κ_{ij} merely raises or lowers the calculated pressure.

Computational Procedure

To compute the fugacity of a component i in a one-phase mixture, we use eq 9 to 14 and A7. To do so, we require the temperature, composition, and volume of that mixture. Normally, the volume is not known and must be found (by trial and error) using eq A2 knowing the temperature, pressure, and composition. The required constants are obtained from Table A-I, from pure-component critical constants, and from the mixing rules given in eq 9 to 14. For mixtures containing hydrogen and helium we use slightly temperature-dependent effective critical constants as discussed elsewhere (Gunn et al., 1966).

For phase-equilibrium calculations the fugacity of every component i must be found in each phase (see eq 4). The calculational procedure for the gas phase is identical with that for the liquid phase. The essential elements of the procedure are summarized in Table A-II.

From a computational point of view, the most time-consuming procedure is the calculation of the liquid volume and the vapor volume. An efficient method for performing this calculation has been presented previously by Plöcker and Knapp (1976). While this previous publication refers to the

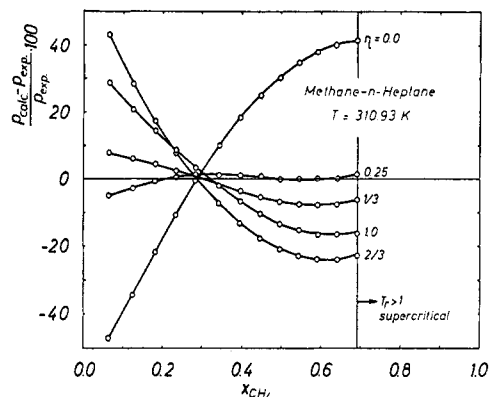


Figure 1. Deviations between calculated and experimental bubble pressures for the binary system methane-*n*-heptane at 310.93 K for various values of η . Parameter κ_{ij} is optimized for any value of η using the experimental data of Reamer et al. (1956).

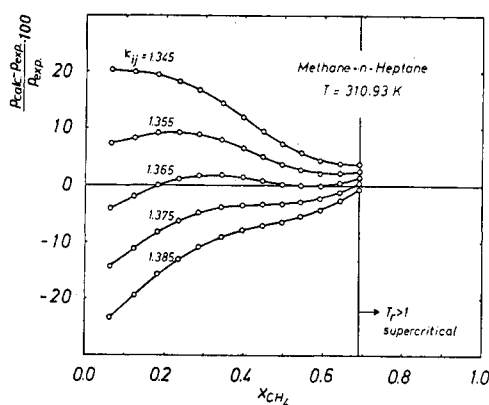


Figure 2. Deviations between calculated and experimental bubble pressures for the binary system methane-*n*-heptane at 310.93 K for various values of κ_{ij} ; $\eta = 0.25$; experimental data from Reamer et al. (1956).

original Benedict-Webb-Rubin equation and to its modification by Starling (1973), it is easily applied to the Lee-Kesler equation used here.

Results

Vapor-liquid equilibrium calculations were performed for over 100 binary systems using data taken from the literature. To a good approximation, binary parameters κ_{ij} were found to be independent of temperature, density, and composition. Table I presents κ_{ij} values for about 150 binary systems. To calculate vapor-liquid equilibria in ternary (or higher) systems, no ternary (or higher) parameters are required. Table I also includes a few results for polar mixtures (e.g., water and ammonia). One should not, however, conclude that the procedure described here is generally applicable to such mixtures. Rather it appears fortuitous that good results were obtained for the particular polar mixtures included in Table I. Generally, one should not expect such fortunate results since strongly polar fluids often show significant deviations from corresponding-states behavior. The inclusion of polar mixtures within a generalized corresponding-states phase equilibrium procedure is still in a preliminary stage.

Some typical results, especially for strongly asymmetric mixtures, are shown in Figures 3 to 7. Calculated K factors ($K = y/x$) are in good agreement with experiment, especially if one considers that in some cases (especially when $K \ll 1$) appreciable experimental uncertainty is likely. The calculational procedure described here is not restricted to low or in-

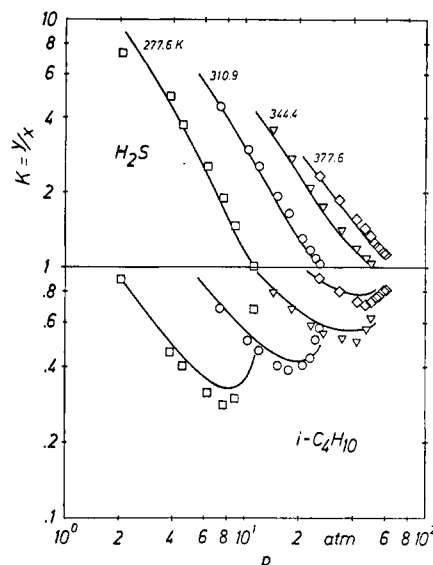


Figure 3. Calculated K factors for the binary system hydrogen sulfide-isobutane; comparison with experimental data of Robinson and Besserer (1972).

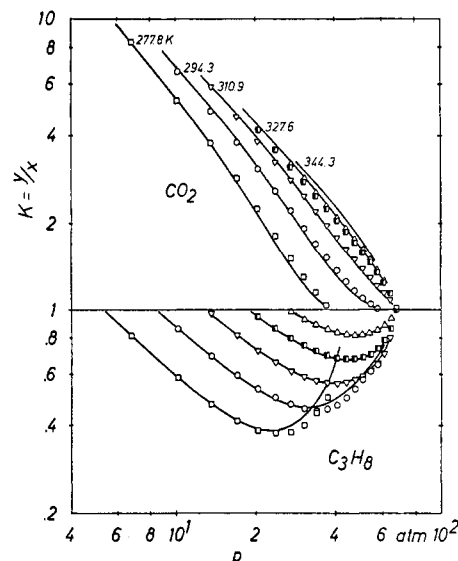


Figure 4. Calculated K factors for the binary system carbon dioxide-propane; comparison with experimental data of Reamer et al. (1951).

intermediate pressures but is applicable also to high pressures, close to the critical region. Calculations in the critical region (approximately $T/T_c(\text{true}) > 0.96$) are discussed briefly in Appendix B.

Figures 8 and 9 present results for light gases in high boiling solvents. It is gratifying that the calculated Henry's constants are not only in good agreement with experiment but also give a good representation of the effect of temperature.

Correlation of Binary Parameters κ_{ij}

Although Table I gives parameters for a large number of binary systems, it is desirable to estimate such parameters for other binaries where insufficient experimental information is at hand. Figures 10 to 13 present κ_{ij} parameters as a function of the dimensionless group $(T_{ci} \cdot v_{ci})(T_{cj} \cdot v_{cj})^{-1}$.

Figure 10 gives results for mixtures of hydrocarbons with hydrocarbons (including aromatics), Figure 11 for mixtures of nitrogen with hydrocarbons, Figure 12 for mixtures of carbon dioxide with hydrocarbons, and Figure 13 for mixtures of hydrogen with hydrocarbons. While the experimental evi-

Table I. Binary Parameters κ_{ij}

System	κ_{ij}	System	κ_{ij}
Methane-ethane	1.052	Benzene-cyclohexane	0.979
-ethylene	1.014	- <i>n</i> -heptane	0.985
-propane	1.113	- <i>n</i> -octane	0.987
-propylene	1.089	-isooctane	0.982
- <i>n</i> -butane	1.171	- <i>n</i> -nonane	1.034
-isobutane	1.155	- <i>n</i> -decane	1.047
- <i>n</i> -pentane	1.240	Cyclohexane- <i>n</i> -heptane	0.999
-isopentane	1.228	- <i>n</i> -octane	1.010
- <i>n</i> -hexane	1.304	- <i>n</i> -nonane	1.021
-cyclohexane	1.269	- <i>n</i> -decane	1.032
-benzene	1.234	<i>n</i> -Heptane- <i>n</i> -octane	0.993
- <i>n</i> -heptane	1.367	-isooctane	1.002
- <i>n</i> -octane	1.423	- <i>n</i> -nonane	1.002
- <i>n</i> -nonane	1.484	- <i>n</i> -decane	1.010
- <i>n</i> -decane	1.533	<i>n</i> -Octane- <i>n</i> -nonane	0.993
Ethane-ethylene	0.991	- <i>n</i> -decane	0.999
-propane	1.010	<i>n</i> -Nonane- <i>n</i> -decane	0.991
-propylene	1.002	Nitrogen-methane	0.977
- <i>n</i> -butane	1.029	-ethylene	1.032
-isobutane	1.036	-ethane	1.082
- <i>n</i> -pentane	1.064	-propane	1.177
-isopentane	1.070	-propylene	1.151
- <i>n</i> -hexane	1.106	- <i>n</i> -butane	1.276
-cyclohexane	1.081	- <i>n</i> -pentane	1.372
-benzene	1.066	- <i>n</i> -hexane	1.442
- <i>n</i> -heptane	1.143	-oxygen	0.997
- <i>n</i> -octane	1.165	-carbon monoxide	0.987
- <i>n</i> -nonane	1.214	-argon	0.988
- <i>n</i> -decane	1.237	-hydrogen sulfide	0.983
Ethylene- <i>n</i> -butane	0.998	-carbon dioxide	1.110
-benzene	1.094	-nitrous oxide	1.073
- <i>n</i> -heptane	1.163	-ammonia	1.033
Acetylene-ethylene	0.948	Carbon dioxide-methane	0.975
Propane-propylene	0.992	-ethane	0.938
- <i>n</i> -butane	1.003	-propane	0.925
-isobutane	1.003	- <i>n</i> -butane	0.955
- <i>n</i> -pentane	1.006	-isobutane	0.946
-isopentane	1.009	- <i>n</i> -pentane	1.002
- <i>n</i> -hexane	1.047	- <i>n</i> -hexane	1.018
-cyclohexane	1.037	-cyclohexane	1.054
-benzene	1.011	-benzene	1.018
- <i>n</i> -heptane	1.067	- <i>n</i> -heptane	1.058
- <i>n</i> -octane	1.090	- <i>n</i> -octane	1.090
- <i>n</i> -nonane	1.115	- <i>n</i> -nonane	1.126
- <i>n</i> -decane	1.139	- <i>n</i> -decane	1.160
Propylene- <i>n</i> -butane	1.010	-hydrogen sulfide	0.922
-isobutane	1.009	-R-12	0.969
-isobutene	1.006	-methanol	1.069
<i>n</i> -Butane-isobutane	1.001	Hydrogen-methane	1.216
- <i>n</i> -pentane	0.994	-ethane	1.604
-isopentane	0.998	-ethylene	1.498
- <i>n</i> -hexane	1.018	-propane	1.826
-cyclohexane	1.008	- <i>n</i> -butane	2.093
-benzene	0.999	- <i>n</i> -pentane	2.335
- <i>n</i> -heptane	1.027	- <i>n</i> -hexane	2.456
- <i>n</i> -octane	1.046	- <i>n</i> -heptane	2.634
- <i>n</i> -nonane	1.064	-nitrogen	1.080
- <i>n</i> -decane	1.078	-carbon monoxide	1.085
<i>n</i> -Pentane-isopentane	0.987	-carbon dioxide	1.624
- <i>n</i> -hexane	0.996	Argon-oxygen	0.985
-cyclohexane	0.996	-ammonia	1.010
-benzene	0.977	-methane	0.984
- <i>n</i> -heptane	1.004	Oxygen-nitrous oxide	1.057
- <i>n</i> -octane	1.020	Carbon monoxide-methane	0.974
- <i>n</i> -nonane	1.033	Krypton-oxygen	0.989
- <i>n</i> -decane	1.045	Hydrogen sulfide-isobutane	0.947
<i>n</i> -Hexane-cyclohexane	0.998	Nitrous oxide-methane	1.017
-benzene	0.978	Water-carbon dioxide	0.920 ^a
- <i>n</i> -heptane	1.008	-ammonia	1.152 ^a
- <i>n</i> -octane	1.005	-methanol	0.979 ^a
- <i>n</i> -nonane	1.015		
- <i>n</i> -decane	1.025		

^a See comment under Results.

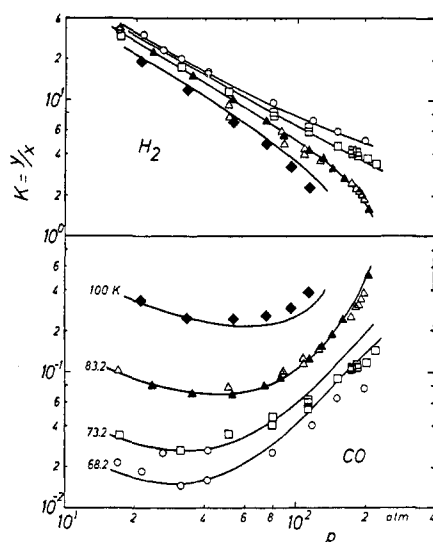


Figure 5. Calculated K factors for the binary system hydrogen-carbon monoxide; comparison with experimental data of Verschoye (1931) ($\square \triangle$) and Akers and Eubanks (1960) ($\blacktriangle \blacklozenge$).

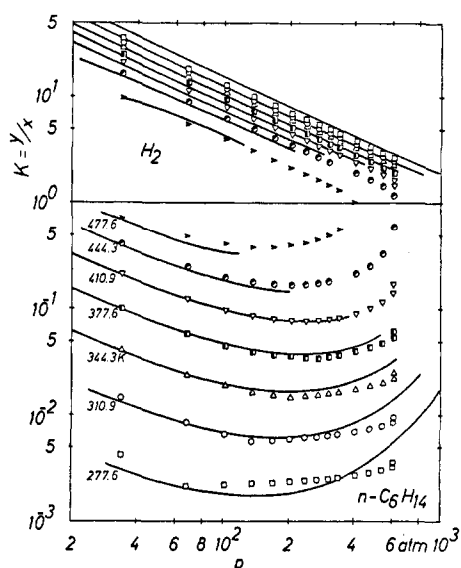


Figure 6. Calculated K factors for the binary system hydrogen- n -hexane; comparison with experimental data of Nichols et al. (1957).

dence supporting Figure 10 is strong, that in support of the other three is somewhat weaker and will require modification as new experimental data become available.

Figures 10 to 13 indicate that in almost all cases κ_{ij} is different from unity; even a rough estimation of κ_{ij} is usually better than setting $\kappa_{ij} = 1$. However, Figures 11 to 13 must be used with caution and should always yield priority to reliable experimental information, even when that information is of limited quantity.

Figure 12 often provides reasonable estimates for systems containing hydrogen sulfide and hydrocarbons because the weak dipolar nature of H_2S is roughly equivalent to the weak quadrupolar nature of CO_2 . To illustrate, consider the quaternary system methane(1)-carbon dioxide(2)-hydrogen sulfide(3)-normal octane(4). All values of κ_{ij} (except κ_{13} and κ_{34}) were obtained from Table I. For the hydrogen sulfide-hydrocarbon parameters κ_{13} and κ_{34} we chose the same values as for the carbon dioxide-hydrocarbon parameters κ_{12} and κ_{24} . Calculated results at 273.15 K are in good agreement with

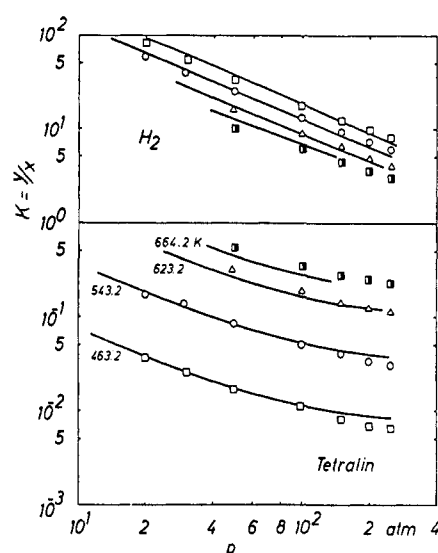


Figure 7. Calculated K factors for the binary system hydrogen-tetralin; comparison with experimental data of Simnick and Chao (1978).

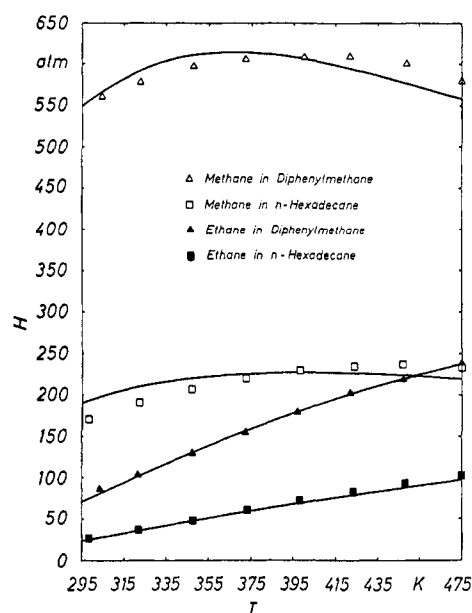


Figure 8. Calculated Henry's constants for methane and ethane in diphenylmethane and n -hexadecane; comparison with experimental data of Cukor and Prausnitz (1972).

experimental measurements of Mundis et al. (1977) as shown in Figure 14.

The method presented here is in some respect similar to that of Starling (1973) in the sense that both methods use a modified form of the Benedict-Webb-Rubin equation and require critical data and acentric factors as primary input. For fifteen representative binary systems, Table II compares deviation in calculated and experimental pressure and vapor composition between those obtained by Starling (A) and those obtained by the technique used here (B). For symmetric (or nearly symmetric) systems, the results do not differ appreciably. However, for highly asymmetric systems, the new method is superior. It is likely that this superiority is a direct consequence of the new mixing rule, eq 9.

Enthalpy

The calculational procedure described in this work also provides residual enthalpies, as discussed in Appendix A.

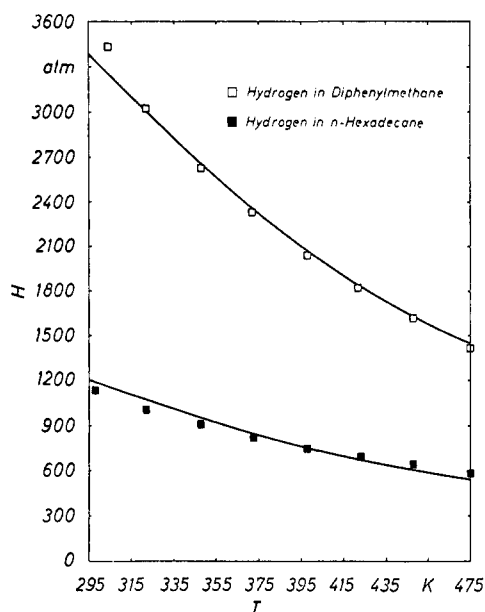


Figure 9. Calculated Henry's constants for hydrogen in diphenylmethane and *n*-hexadecane; comparison with experimental data of Cukor and Prausnitz (1972).

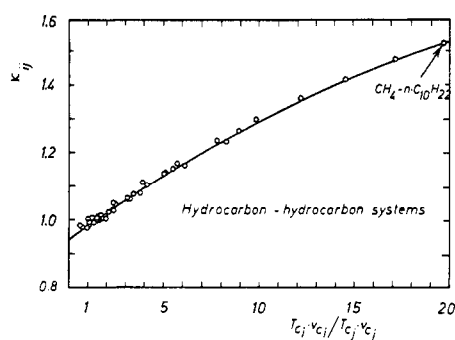


Figure 10. Correlation of binary parameters, κ_{ij} for hydrocarbon-hydrocarbon systems, including aromatics.

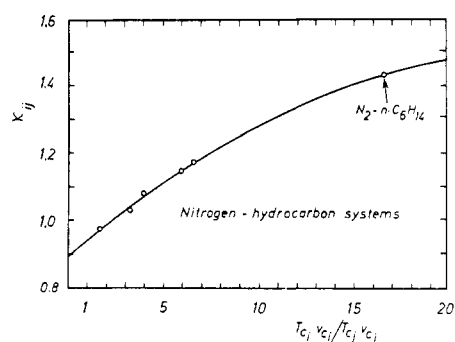


Figure 11. Correlation of binary parameters κ_{ij} for nitrogen-hydrocarbon systems.

Table III compares calculated and experimental results for seven binary mixtures including both single phases and the two-phase region. For calculation in the two-phase region it is necessary to know the K factors. These were not taken from the experimental data but were calculated from a flash program using the method described here. Therefore, since two-phase enthalpies are very sensitive to K factors, agreement between experiment and calculation, while excellent in the one-phase region, is somewhat lower in the two-phase region. The results shown in Table III are similar to those

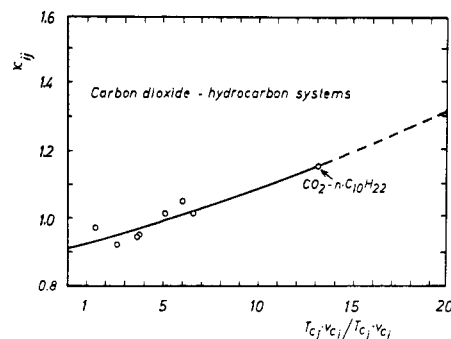


Figure 12. Correlation of binary parameters κ_{ij} for carbon dioxide-hydrocarbon systems.

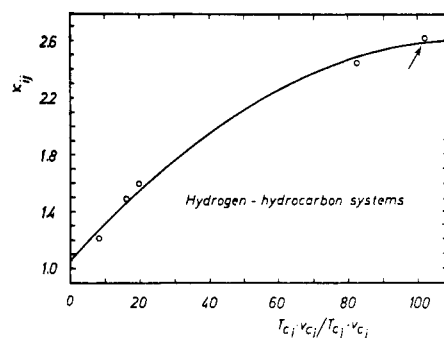


Figure 13. Correlation of binary parameters κ_{ij} for hydrogen-hydrocarbon systems. The arrow indicates data for the H_2 - C_7H_{16} system.

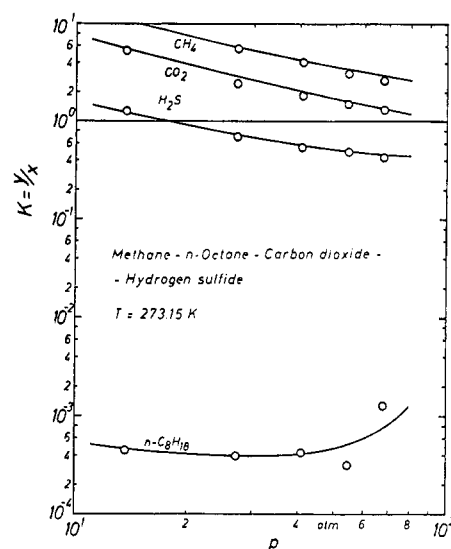


Figure 14. Calculated K factors for the quaternary system methane-*n*-octane-carbon dioxide-hydrogen sulfide; comparison with experimental data of Mundis et al. (1977).

based on Starling's equation but are superior to those obtained from most other common methods for the one-phase region as indicated by Tarakad and Danner (1976).

Computer Calculations

The necessary computer programs were written in Fortran IV. For a typical binary system, calculations of K factors and enthalpies require approximately 0.1 to 0.2 s on a CDC 6500 machine. Documented programs including computer flow sheets are available upon request to the authors.

Table II. Comparison of Experimental and Calculated Vapor-Liquid Equilibria Using the Methods of BWR-Stirling (A) and This Work (B)

Components 1 2		Experimental data			Deviations (experimental - calculated)							
		Ref	Temp region, K	Max. press., atm	Pressure, %				Vapor comp., mol%			
					Average		RMS ^b		Average		RMS ^b	
					A	B	A	B	A	B	A	B
CH ₄ -C ₂ H ₆		c,d,e	130-284	55	3.0	1.3	3.6	1.4	0.6	0.4	0.9	0.4
CH ₄ -C ₃ H ₈		f,g	130-361	90	7.9	1.9	5.0	1.8	1.1	0.8	1.6	0.5
CH ₄ -n-C ₅ H ₁₂		h,i	277-460	150	11.6	1.5	8.2	1.6	1.7	2.1	1.8	1.6
CH ₄ -n-C ₇ H ₁₆		j,k	199-511	105	27.7	5.2	14.5	3.5	1.3	0.8	1.6	1.0
C ₂ H ₆ -n-C ₆ H ₁₄		l	273-374	52	5.8	2.6	3.9	2.6	0.7	0.6	0.8	0.7
C ₃ H ₈ -n-C ₁₀ H ₂₂		M	277-511	55	22.0	4.4	11.3	5.8	1.6	0.6	1.9	0.9
C ₂ H ₆ -C ₆ H ₆ ^a		n	273-534	82	5.1	2.0	5.0	1.4	1.4	1.6	3.2	1.3
C ₂ H ₆ -c-C ₆ H ₁₂ ^a		o	233-534	85	6.3	2.5	3.6	2.3	2.1	1.5	1.6	1.4
C ₆ H ₆ -n-C ₇ H ₁₆ ^a		p,q,r	333-372	1	1.0	0.9	0.5	0.7	0.6	0.5	0.3	0.3
N ₂ -CH ₄		s,t,u	88-180	48	4.4	0.7	2.3	0.6	1.3	0.6	0.9	0.5
N ₂ -C ₃ H ₈		v,w	143-354	211	22.0	4.2	19.4	4.2	2.0	1.5	1.6	1.9
N ₂ -H ₂ S		x	256-345	205	13.4	5.2	12.6	4.3	3.4	2.6	2.7	2.3
CO ₂ -CH ₄		y,z	173-271	76	4.9	2.6	3.8	2.0	1.4	0.7	1.2	0.7
CO ₂ -C ₃ H ₈		aa,bb, cc,dd	233-360	61	11.8	3.6	6.6	2.2	4.9	0.9	2.0	0.7
H ₂ S-CO ₂		ee,ff	224-360	80	9.7	1.9	7.6	1.4	2.3	1.0	1.2	0.8

^a Special adjustment of BWR-Stirling's binary parameter k_{ij} was necessary. For all other systems the original proposed values were used. ^b RMS: root-mean-square deviation. ^c Price, A. R., Thesis, Rice Institute, Houston, Texas, 1957. ^d Wichterle, I., Kobayashi, R., *J. Chem. Eng. Data*, 17, (1), 9 (1972). ^e Wichterle, I., Kobayashi, R., *J. Chem. Eng. Data*, 17 (1), 13 (1972). ^f Reamer, H. H., Sage, B. H., Lacey, W. N., *Ind. Eng. Chem.*, 42, (3), 534 (1950). ^g Wichterle, I., Kobayashi, R., *J. Chem. Eng. Data*, 17 (1), 4 (1972). ^h Berry, V. M., Sage, B. H., NSRDS-NBS, 32 (1970). ⁱ Sage, B. H., Reamer, H. H., Olds, R. H., Lacey, W. N., *Ind. Eng. Chem.*, 34 (9), 1108 (1942). ^j Chang, H. L., Hurt, L. J., Kobayashi, R., *AIChE J.*, 12 (6), 1212 (1966). ^k Reamer, H. H., Sage, B. H., Lacey, W. N., *Ind. Eng. Chem. Data Ser.*, 1 (1), 29 (1956). ^l Rodrigues, A. B. J., McCaffrey, D. S., Jr., Kohn, J. P., *J. Chem. Eng. Data*, 13 (2), 165 (1968). ^m Reamer, H. H., Sage, B. H., *J. Chem. Eng. Data*, 11 (1), 17 (1966). ⁿ Kay, W. B., Nevens, T. D., *Chem. Eng. Prog. Sym. Ser.*, 48, (3), 108 (1952). ^o Kay, W. B., Albert, R. E., *Ind. Eng. Chem.*, 48 (3), 422 (1956). ^p Hanson, D. O., van Winkle, M., *J. Chem. Eng. Data*, 12 (3), 319 (1967). ^q Myers, H. S., *Ind. Eng. Chem.*, 47, (10), 2215 (1955). ^r Sieg, L., *Chem. Ing. Techn.*, 22 (15), 323 (1950). ^s Bloomer, O. T., Parent, J. D., *Chem. Eng. Prog. Symp. Ser.*, 49 (3), 11 (1953). ^t Kidnay, A. J., Miller, R. C., Parrish, W. R., Hiza, M. J., *Cryogenics*, 531 (1975). ^u Parrish, W. R., Hiza, M. J., Cryogenic Engineering Conference, Atlanta, Ga., Paper K-9, 1973. ^v Grauso, L., Fredenslund, A., Mollerup, J., to be published, 1978. ^w Schindler, D. L., Swift, G. W., Kurata, F., *Hydrocarbon Process.*, 45 (11), 205 (1966). ^x Robinson, D. B., Besserer, G. J., *NGPA Res. Rep.*, 7 (1972). ^y Donnelly, H. G., Katz, D. L., *Ind. Eng. Chem.*, 46, (1), 511 (1954). ^z Neumann, A., Walch, W., *Chem. Ing. Techn.*, 40 (5), 241 (1968). ^{aa} Akers, W. W., Kelley, R. E., Lipscomb, T. G., *Ind. Eng. Chem.*, 46 (12), 2535 (1954). ^{bb} Haman, S. E. M., Lu, B. C.-Y., *J. Chem. Eng. Data*, 21 (2), 200 (1976). ^{cc} Poettmann, F. H., Katz, D. L., *Ind. Eng. Chem.*, 37 (9), 847 (1945). ^{dd} Reamer, H. H., Sage, B. H., Lacey, W. N., *Ind. Eng. Chem.*, 43 (11), 2515 (1951). ^{ee} Bierlein, J. A., Kay, W. B., *Ind. Eng. Chem.*, 45 (3), 618 (1953). ^{ff} Sobocinsky, D. P., Kurata, F., *AIChE J.*, 5 (4), 545 (1959).

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Appendix A

Summary of the Main Equations for the Calculation of Thermodynamic Properties. Lee and Kesler use eq 1 for calculating the compressibility factor (and derived quantities) for the fluid of interest

$$z = z^{(0)} + \frac{\omega}{\omega(r)} \cdot (z^{(r)} - z^{(0)}) \quad (\text{A1})$$

To calculate the compressibility factor of a simple fluid $z^{(0)}$ and of a reference fluid $z^{(r)}$, the same modification of the Benedict-Webb-Rubin equation is used

$$z = \frac{p_r v_r}{T_r} = 1 + \frac{B}{v_r} + \frac{C}{v_r^2} + \frac{D}{v_r^5} + \frac{c_4}{T_r^3 v_r^2} \left(\beta + \frac{\gamma}{v_r^2} \right) \exp \left(-\frac{\gamma}{v_r^2} \right) \quad (\text{A2})$$

with

$$B = b_1 - b_2/T_r - b_3/T_r^2 - b_4/T_r^3$$

$$C = c_1 - c_2/T_r + c_3/T_r^3$$

$$D = d_1 + d_2/T_r$$

There are two universal sets of constants for $b_1, b_2, b_3, b_4, c_1, c_2, c_3, d_1, d_2, \beta$, and γ , one for the simple fluid, and one for the reference fluid. They are given in Table A-I. From eq A2 we find the derived thermodynamic quantities.

Fugacity Coefficient ϕ .

$$\ln \phi = z - 1 - \ln z + \frac{B}{v_r} + \frac{C}{2v_r^2} + \frac{D}{5v_r^5} + E \quad (\text{A3})$$

with

$$E = \frac{c_4}{2T_r^3 \gamma} \left[\beta + 1 - \left(\beta + 1 + \frac{\gamma}{v_r^2} \right) \exp \left(-\frac{\gamma}{v_r^2} \right) \right]$$

Isothermal Enthalpy Departure.

$$\frac{\Delta h}{R \cdot T_c} = \frac{h - h^0}{R \cdot T_c} = T_r \left[z - 1 - \frac{b_2 + 2b_3/T_r + 3b_4/T_r^2}{T_r \cdot v_r} - \frac{c_2 - 3c_3/T_r^2}{2T_r v_r^2} + \frac{d_2}{5T_r v_r^5} + 3E \right] \quad (\text{A4})$$

Isothermal Entropy Departure.

$$\frac{\Delta s}{R} + \ln \frac{p}{p^0} = \frac{s - s^0}{R} + \ln \frac{p}{p^0} = \ln z - \frac{b_1 + b_3/T_r^2 + 2b_4/T_r^3}{v_r} - \frac{c_1 - 2c_3/T_r^3}{2v_r^2} - \frac{d_1}{5v_r^5} + 2E \quad (\text{A5})$$

The equations for isochoric and isobaric heat capacity departures are given by Lee and Kesler (1975).

Table A-II presents the procedure for calculation of any of the derived thermodynamic quantities from known values of

Table III. Comparison of Experimental and Calculated Isothermal Enthalpy Departures Using the Methods of BWR–Starling (A) and This Work (B)

Components			Experimental data			Total			Two-phase region				Liquid-phase region				Vapor-phase/supercrit. reg.						
			Ref	Temp region, K	Max. press., atm	No. of data prints	A	J/g	%	B	J/g	%	No. of data prints	A	J/g	%	No. of data prints	A	J/g	%			
1	2																						
CH ₄ -C ₃ H ₈ C ₃ H ₈ -C ₃ H ₆ N ₂ -CH ₄ n-C ₅ H ₁₂ -n-C ₈ H ₁₈ -C ₆ H ₁₄ -C ₈ H ₁₈ C ₆ H ₆ -n-C ₈ H ₁₈	a, b, c	99-422	136	1827	7.7	3.3	4.7	2.5	180	31.8	15.7	21.1	10.0	824	7.8	1.8	3.9	0.9	823	3.2	2.6	1.9	2.4
	d	99-422	136	199	5.3	2.0	3.0	1.5	4	9.5	5.5	5.7	4.0	117	6.3	1.5	2.9	0.7	78	3.2	2.9	3.1	2.6
	c	99-422	136	462	1.5	1.5	1.8	4.0	7	4.8	2.5	12.7	8.0	80	4.5	1.6	1.8	0.6	375	0.7	1.4	1.6	4.6
	g	297-589	95	350	9.7	8.9	5.5	7.1	13	27.9	16.6	9.9	6.0	155	7.8	3.8	2.7	1.2	182	11.4	16.2	7.5	11.3
	h	399-633	95	546	4.2	15.0	3.6	16.6	6	10.7	7.8	9.9	16.0	140	6.2	2.4	2.8	1.2	400	3.2	21.1	3.7	22.4
	f	422-644	95	534	7.4	9.5	5.4	8.9	8	49.3	29.9	25.2	13.9	88	12.1	4.8	6.5	2.5	438	4.8	10.8	4.7	10.1
e	466-589	95	345	8.4	8.9	6.0	8.3	9	68.1	42.2	37.7	23.0	144	9.9	5.8	6.0	4.1	192	4.2	10.0	3.1	9.1	
Total				4236	6.7	6.7	4.4	6.0	227	31.8	16.5	20.4	10.4	1548	7.8	2.6	4.8	1.3	2488	3.8	8.4	2.9	8.5
<i>a</i> Yesavage, V. F., Katz, D. L., Powers, J. E., <i>J. Chem. Eng. Data</i> , 14 (2), 137 (1969). <i>b</i> Yesavage, V. F., Katz, D. L., Powers, J. E., <i>J. Chem. Eng. Data</i> , 16 (5), 867 (1970). <i>c</i> Mather, A. E., Ph.D. Thesis, University of Michigan, 1967. <i>d</i> Furtado, A. W., Ph.D. Thesis, University of Michigan, 1973. <i>e</i> Lenoir, J. M., Hayworth, K. E., Hipkin, H. G., <i>J. Chem. Eng. Data</i> , 16 (3), 280 (1971). <i>f</i> Lenoir, J. M., Hipkin, H. G., <i>J. Chem. Eng. Data</i> , 17 (3), 319 (1972). <i>g</i> Lenoir, J. M., Robinson, D. R., Hipkin, H. G., <i>J. Chem. Eng. Data</i> , 15 (1), 166 (1970). <i>h</i> Lenoir, J. M., Kuravila, G. K., Hipkin, H. G., <i>J. Chem. Eng. Data</i> , 16 (3), 271 (1971).																							

a Yesavage, V. F., Katz, D. L., Powers, J. E., *J. Chem. Eng. Data*, 14 (2), 137 (1969). b Yesavage, V. F., Katz, D. L., Powers, J. E., *J. Chem. Eng. Data*, 16 (5), 867 (1970).

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Table A-I. Constants for the BWR–Lee–Kesler Equation

Con- stant	Simple fluid	Reference fluid	Con- stant	Simple fluid	Reference fluid
b_1	0.1181193	0.2026579	c_3	0.0	0.016901
b_2	0.265728	0.331511	c_4	0.042724	0.041577
b_3	0.154790	0.027655	$d_1 \times 10^4$	0.155428	0.48736
b_4	0.030323	0.203488	$d_2 \times 10^4$	0.623689	0.0740336
c_1	0.0236744	0.0313385	β	0.65392	1.226
c_2	0.0186984	0.0503618	γ	0.060167	0.03754
$\omega^{(r)} = 0.3978$.					

Table A-II. Calculation Procedure^a

Step	From	Calculate
1	T, p, T_c, p_c	T_r, p_r
2	T_r, p_r , eq A2 with constants "simple fluid"	$v_r^{(0)}, z^{(0)}, (\ln \phi)^{(0)}, (\Delta h/RT_c)^{(0)}$, etc.
3	T_r, p_r , eq A2 with constants "reference fluid"	$v_r^{(r)}, z^{(r)}, (\ln \phi)^{(r)}, (\Delta h/RT_c)^{(r)}$, etc.
4	$q = q^{(0)} + \omega/\omega^{(r)} \cdot (q^{(r)} - q^{(0)})$	$q = z, \ln \phi, \Delta h/RT_c$, etc.
	$q^{(0)} = z^{(0)}, (\ln \phi)^{(0)}$, etc.	
	$q^{(r)} = z^{(r)}, (\ln \phi)^{(r)}$, etc.	

^a For mixtures use T_{cm}, p_{cm}, ω_M instead of T_c, p_c, ω .

temperature and pressure; the parameters T_c, p_c , and ω for each pure fluid, and the pseudo-criticals T_{cm}, p_{cm} , and ω_M for mixtures.

The fugacity coefficient for component i in a mixture can be found by differentiating ϕ_M as shown in eq 6. For the three-parameter corresponding-states principle as used here

$$\ln \phi_M = f(T_r, p_r, \omega) = f(T/T_{cm}, p/p_{cm}, \omega_M) \quad (A6)$$

we obtain (see, e.g., Prausnitz, 1968)

$$\ln \phi_i = \ln \phi_M - \frac{1}{T R \cdot T_{cm}} \sum_{j \neq i} Z_j \left(\frac{dT_{cm}}{dZ_j} \right)_{Z_k} + \frac{z_M - 1}{p_{cm}} \sum_{j \neq i} Z_j \left(\frac{dp_{cm}}{dZ_j} \right)_{Z_k} - \left(\frac{\partial \ln \phi_M}{\partial \omega_M} \right)_{T_r, p_r} \sum_{j \neq i} Z_j \left(\frac{d\omega_M}{dZ_j} \right)_{Z_k} \quad (k \neq i, j) \quad (A7)$$

with

$$\left(\frac{\partial \ln \phi_M}{\partial \omega_M} \right)_{T_r, p_r} = \frac{1}{\omega^{(r)}} [(\ln \phi_M)^{(r)} - (\ln \phi_M)^{(0)}] \quad (A8)$$

For $\Delta h_M/RT_{cm}$ and $\ln \phi_M$ we use the procedure described in Table A-II. The derivatives of the pseudo criticals, defined by the mixing rules in eq 9 to 14, are

$$\left(\frac{dT_{cm}}{dZ_j} \right)_{Z_k} = \left[2 \sum_l Z_l (v_{clj} \eta T_{clj} - v_{cli} \eta T_{cli}) - \eta v_{cm} \eta^{-1} \left(\frac{dv_{cm}}{dZ_j} \right)_{Z_k} T_{cm} \right] / v_{cm} \eta \quad (A9)$$

$$\left(\frac{dv_{cm}}{dZ_j} \right)_{Z_k} = 2 \sum_l Z_l (v_{clj} - v_{cli}) \quad (A10)$$

$$\left(\frac{dp_{cm}}{dZ_j} \right)_{Z_k} = p_{cm} \left[\left(\frac{dz_{cm}}{dZ_j} \right)_{Z_k} / z_{cm} + \left(\frac{dT_{cm}}{dZ_j} \right)_{Z_k} / T_{cm} - \left(\frac{dv_{cm}}{dZ_j} \right)_{Z_k} / v_{cm} \right] \quad (A11)$$

$$\left(\frac{dz_{cm}}{dZ_j} \right)_{Z_k} = -0.085 \left(\frac{d\omega_M}{dZ_j} \right)_{Z_k} \quad (A12)$$

$$\left(\frac{d\omega_M}{dZ_j} \right)_{Z_k} = \omega_j - \omega_i \quad (A13)$$

In eq A9–A13, $k \neq i, j$.

Appendix B

Calculations in the Critical Region. Since the pseudo-critical constants are empirical correlating parameters normally not identical with the true critical constants, problems arise for calculations in the critical region. In most cases we find that the pseudo-critical temperature T_{cm} of the mixture is lower than the true critical temperature T_{ct} . Prausnitz and Chueh (1968) proposed a correction of T_{cm} for reduced temperatures $0.93 < T/T_{cm} < 1.0$ such that $T_{cm} = T_{ct}$ when the true critical point is reached. This method, first applied to the calculation of the liquid densities using the correlation of Lyckman et al. (1965) was also applied to the method described here. Results, however, were not encouraging. The problem of calculating K factors in the critical region, therefore, remains. However, our experience shows that in most cases no problems appear for $T/T_{ct} < 0.96$.

Nomenclature

B, C, D = abbreviations in the BWR–Lee–Kesler equation for functions depending on temperature only
 $b_1, b_2, b_3, b_4, c_1, c_2, c_3, d_1, d_2, \beta, \gamma$ = constants in the BWR–Lee–Kesler equation
 f = fugacity
 h = molar enthalpy
 Δh = isothermal enthalpy departure
 H = Henry's constant
 K = y/x
 n = number of moles
 p = pressure
 R = universal gas constant
 s = molar entropy
 Δs = isothermal entropy departure
 T = temperature
 v = molar volume
 x = mole fraction, liquid phase
 y = mole fraction, vapor phase
 z = compressibility factor
 Z = mole fraction

Greek Letters

η = universal exponent in the mixing rule for the pseudo-critical temperature
 κ_{ij} = binary parameter for the i – j mixture in the mixing rule for the pseudo-critical temperature
 ϕ = fugacity coefficient
 ω = acentric factor

Subscripts

c = critical property
 i, j, k = component i, j, k
 M = mixture property; here: pseudo-critical property
 r = reduced property

Superscripts

L = liquid phase
 V = vapor phase
 0 = standard state
 (0) = simple fluid
 (r) = reference fluid

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